

Yaw Stability Control By LPV Strategy

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Abstract

This work is oriented to the modelling and control of the lateral dynamic of a vehicle. Much effort is dedicated to the description of the dynamic model chosen, in terms of equations which characterize the evolution of some variables of interest during cornering manoeuvres. Second phase introduces to the control law adopted, H_∞ LPV approach, analyzing the performance obtained also under the action of disturbance effect.

1 Introduction

Automotive control design is mainly oriented to develop intelligent systems capable of improving the driving experience in terms of comfort and safety. In recent years, most of the research effort has been devoted to introduce assistance systems for reducing accident risk, in most of the cases due to distraction of the drivers, and also to make the driving experience the best as possible. Originally developed for trucks and sport cars, **Advanced Driving Assistance Systems (ADAS)** are rapidly becoming an integrating part of a vehicle, thus the need of developing algorithms (controllers) which thanks to sensors measurements are in charge of making intervention together or in lieu of the driver. Based on the grade of decision autonomy, autonomous vehicles are classified in the following manner:

0. **No Automation:** The driving tasks (steering, braking etc.) are all performed by human.
1. **Driver Assistance:** The vehicle features a single automated system (cruise control for instance).
2. **Partial Automation:** The vehicle can perform steering and acceleration tasks even if the human can take the control at any time.
3. **Conditional Automation:** The vehicle, thanks to environmental detection, can perform many tasks but human override is still required.
4. **High Automation:** Under specific conditions, thanks to the use of GPS technology to create a virtual geographic boundary, the vehicle can perform all the tasks.
5. **Full Automation:** The vehicle performs all the tasks.

Yaw Stability Control also known as **ESP** (Electronic Stability Program) is a fundamental assistance system which has the important role of guaranteeing the stability of the vehicle during slipping conditions due to high speed cornering or low friction coefficient (wet or icy road). In advanced systems it is coupled with a LKS (Lane Keeping System) controller whose role consists of maintaining the vehicle trajectory inside the lane, thus preventing unexpected lane departures. Basically, yaw stability controllers have to prevent from spinning and drifting out the vehicle as shown in the following figure:

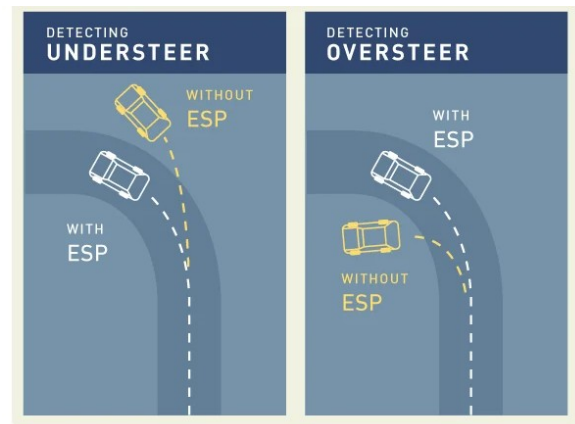


Figure 1: *ESP Representation*

ESP systems aim to guarantee, during critical manoeuvres (high speed cornering or bad weather conditions driving), a yaw rate as close as possible to the one expected from the driver. To do this, they can act in different ways here briefly reported:

- **Differential Braking** systems which utilize the ABS brake system to apply different braking forces between left and right wheels to con-

trol yaw moment.

- **Steer-by-Wire** systems which modify the driver's steering angle input.
- **Active Torque Distribution** systems which independently control the drive torque for each wheel.

Inspired to *Steer-by-Wire* technique, this application wants to design an LPV Controller which regulates the front steering angle to track the desired yaw rate for a selected path.

2 System Model

The scenario considered presumes to obtain a description of the vehicle dynamics during cornering manoeuvres. To this aim is possible to distinguish between high speed and low speed cornering. Indeed, when the velocity of the vehicle is low the directions of travel of the wheels (defined by the velocity vector) are aligned together with the heading direction (orientation of the wheels). When the speed increases, this hypothesis is no longer true and, due to physical reasons, small angles between travel directions and heading directions arise, named **slip angles**. To model this phenomenon, since we are interested in obtaining a controller capable to face with high speed manoeuvres, let's consider the bicycle model illustrated in Figure 2:

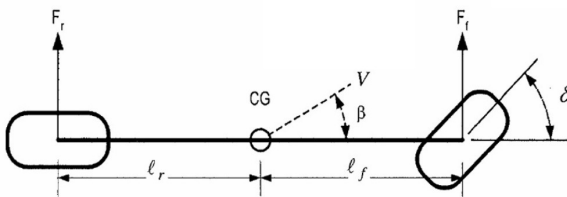


Figure 2: *Bicycle Model*

It embodies the dynamic of the front and rear axle into a unique wheel for each axle (this is possible under the assumption that the left and right front velocities are almost equal, true simplifying the problem) and is characterized by the following variables:

- F_f and F_r are respectively the front and rear centripetal tyre forces.

- l_f and l_r are respectively the front and rear distances between the wheels and the center of gravity CG .
- δ is the front input steering angle.
- v is the velocity vector acting on the CG .
- β is the sideslip angle. It represents the angular offset between the velocity vector and the heading direction of the vehicle.

To obtain the dynamic evolution let's analyse the lateral acceleration of the vehicle. Figure 3 illustrates the global and the local reference frames:

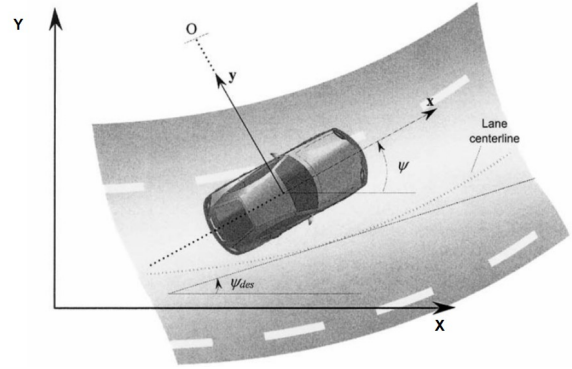


Figure 3: *Global and Local Reference Frames*

The set of coordinates $[X, Y, Z]$ defines the global reference (*inertial system*), while the set $[x, y, z]$ represents the reference frame located on the vehicle CG (*body fixed*) which is rotating with respect to the inertial frame. Consider the origin of the inertial reference frame on the centre of rotation O . The vehicle is travelling along a trajectory defined by the radius of curvature R represented as a vector $\vec{r}$ from O to CG . Applying Merriam and Kraige equation, the acceleration in inertial coordinates can be related to the acceleration in body fixed coordinates as follows:

$$\begin{bmatrix} \ddot{X} \\ \ddot{Y} \\ \ddot{Z} \end{bmatrix} = \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} + \vec{\Omega} \times (\vec{\Omega} \times \vec{r}) + \dot{\vec{\Omega}} \times \vec{r} + 2\vec{\Omega} \times \dot{\vec{r}} \quad (1)$$

Where:

- $\vec{\Omega} = \dot{\psi} \hat{z}$ is the angular velocity of the vehicle perpendicular to the road.

- $\vec{r} = R(-\hat{y})$ is the radius vector which identifies the position of the *CG*. It is oriented in the opposite direction of the lateral local coordinate for geometric reasons.

Substituting in the above equation:

$$\begin{bmatrix} \ddot{X} \\ \ddot{Y} \\ \ddot{Z} \end{bmatrix} = \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} + \dot{\psi} \hat{z} \times (\dot{\psi} \hat{z} \times R(-\hat{y})) + \ddot{\psi} \hat{z} \times R(-\hat{y}) + 2\dot{\psi} \hat{z} \times \dot{R}(-\hat{y}) \quad (2)$$

Solving the vector products:

$$\begin{bmatrix} \ddot{X} \\ \ddot{Y} \\ \ddot{Z} \end{bmatrix} = \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} + \dot{\psi} \hat{z} \times (\dot{\psi} R \hat{x}) + \ddot{\psi} R \hat{x} + 2\dot{\psi} R \hat{x} \quad (3)$$

That is:

$$\begin{bmatrix} \ddot{X} \\ \ddot{Y} \\ \ddot{Z} \end{bmatrix} = \ddot{x} \hat{x} + \ddot{y} \hat{y} + \ddot{z} \hat{z} + \dot{\psi}^2 R \hat{y} + (\ddot{\psi} R + 2\dot{\psi} R) \hat{x} \quad (4)$$

By grouping along the directions:

$$\begin{bmatrix} \ddot{X} \\ \ddot{Y} \\ \ddot{Z} \end{bmatrix} = (\ddot{x} + \ddot{\psi} R + 2\dot{\psi} R) \hat{x} + (\ddot{y} + \dot{\psi}^2 R) \hat{y} + \ddot{z} \hat{z} \quad (5)$$

Thus the inertial lateral acceleration $\ddot{Y} = a_y = \ddot{y} + \dot{\psi}^2 R$. Now, is possible to apply the Dynamic Newton's Second Law along the lateral axis:

$$m a_y = F_f + F_r + F_{bank} \quad (6)$$

With m mass of the vehicle and F_{bank} disturbance force due to the road slope. Thus:

$$m(\ddot{y} + \dot{\psi}^2 R) = F_f + F_r + F_{bank} \quad (7)$$

From the definition of the sideslip angle:

$$\beta = \frac{\dot{y}}{V_x} \implies \dot{y} = \beta V_x \quad (8)$$

Under the hypothesis of constant longitudinal speed during cornering manoeuvre, by differentiating the above equation:

$$\ddot{y} = \dot{\beta} V_x \quad (9)$$

The other term of a_y is:

$$\dot{\psi}^2 R = \dot{\psi} \dot{\psi} R \quad (10)$$

From the definition of angular speed:

$$\dot{\psi} \dot{\psi} R = \dot{\psi} V_x \quad (11)$$

Replacing 11 and 9 in 7 the final result is:

$$m V_x (\dot{\beta} + \dot{\psi}) = F_f + F_r + F_{bank} \quad (12)$$

Now, apply the Dynamic Newton's Second Law along the vertical axis:

$$I_z \ddot{\psi} = l_f F_f - l_r F_r \quad (13)$$

With I_z yaw moment of inertia.

For small slip angles α_f, α_r , the tire forces can be approximated to linear functions of the slip angles:

$$\begin{aligned} F_f &= C_{\alpha f} \alpha_f \\ F_r &= C_{\alpha r} \alpha_r \end{aligned} \quad (14)$$

Looking at Figure 4

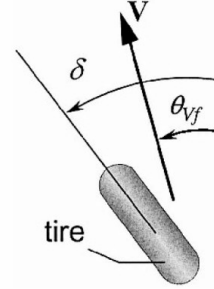


Figure 4: *Tire slip angle*

is possible to compute the slip angles as follows:

$$\begin{aligned} \alpha_f &= \delta - \theta_{vf} \\ \alpha_r &= \delta - \theta_{vr} \end{aligned} \quad (15)$$

Where θ_{vf}, θ_{vr} are the angles between the velocity vector and the longitudinal axis of the vehicle. To compute them, under the assumption of small angles (true since the radius of curvature is large enough), the following relations can be used:

$$\begin{aligned} \theta_{vf} &= \frac{\dot{y} + l_f \dot{\psi}}{V_x} = \beta + \frac{l_f \dot{\psi}}{V_x} \\ \theta_{vr} &= \frac{\dot{y} - l_r \dot{\psi}}{V_x} = \beta - \frac{l_r \dot{\psi}}{V_x} \end{aligned} \quad (16)$$

Substituting 16 in 14 and then in 12 and in 13 the evolutions of the sideslip angle and of the angular acceleration are computed:

$$\begin{aligned}\dot{\beta} &= -\dot{\psi} + \frac{C_{\alpha f}}{mV_x}(\delta - \beta - \frac{l_f \dot{\psi}}{V_x}) + \frac{C_{\alpha r}}{mV_x}(-\beta + \frac{l_r \dot{\psi}}{V_x}) + \frac{g \sin(\theta)}{V_x} \\ \ddot{\psi} &= \frac{l_f C_{\alpha f}}{I_z}(\delta - \beta - \frac{l_f \dot{\psi}}{V_x}) - \frac{l_r C_{\alpha r}}{I_z}(-\beta + \frac{l_r \dot{\psi}}{V_x})\end{aligned}\quad (17)$$

Notice that the above differential equations are linear with respect to $\dot{\psi}$ and β . Thus, recognizing the state and the input of the system as: $x = \begin{bmatrix} \beta \\ \dot{\psi} \end{bmatrix}$, $u = \delta$, the following State-Space representation comes out:

$$\begin{aligned}\dot{x} &= \begin{bmatrix} -\frac{(C_{\alpha f} + C_{\alpha r})}{mV_x} & -\frac{mV_x^2 - C_{\alpha f}l_f + C_{\alpha r}l_r}{mV_x^2} \\ \frac{l_r C_{\alpha r} - l_f C_{\alpha f}}{I_z} & -\frac{l_f^2 C_{\alpha f} - l_r^2 C_{\alpha r}}{I_z V_x} \end{bmatrix} x + \begin{bmatrix} \frac{C_{\alpha f}}{mV_x} \\ \frac{l_f C_{\alpha f}}{I_z} \end{bmatrix} u \\ y &= [0 \quad 1] x + 0 u\end{aligned}\quad (18)$$

For next section, consider the following parameters of the system:

Parameter	Value	Unit
$mass$	1580	Kg
$C_{\alpha f}$	100000	$\frac{N}{rad}$
$C_{\alpha r}$	120000	$\frac{N}{rad}$
l_f	1.3	m
l_r	1.4	m
I_z	2600	$Kg \cdot m^2$

Table 1: *Parameters*

3 Control Design

The yaw stability problem aims to design a controller capable of guaranteeing these features:

- sideslip regulation: the more the sideslip angle is close to zero, the best the stability of the vehicle (no lateral slip of the CG).
- tracking of the yaw rate: during cornering manoeuvres the vehicle should loyally track as much as possible the yaw rate desired $\dot{\psi}_{des} = \frac{V_x}{R}$, which is the desired orientation the vehicle should have when cornering. It accounts the radius of curvature and the current longitudinal velocity, assumed to be constant during the manoeuvre.

Thus, an integral problem is implemented accounting the difference between the desired yaw rate and the current yaw rate of the vehicle. Consequently, the augmented state becomes:

$$\begin{bmatrix} x_e \\ \beta \\ \dot{\psi} \end{bmatrix} \quad (19)$$

Under the hypothesis of measuring at each time instant the current speed of the vehicle, the control design task presumes to develop an H_∞ LPV controller characterized by the following optimization problem:

H_∞ LPV Controller Optimization Problem

Given the optimization problem:

$$\begin{cases} [X^* \ Y_j^*] = \min_{X, Y_j, \gamma} \gamma \\ s.t. \\ \begin{bmatrix} (A_i X + B_{1,i} Y_j) + (A_i X + B_{1,i} Y_j)^T & B_{2,i} & (C_{1,i} X + D_{12,i} Y_j) \\ B_{2,i}^T & -\gamma I_{n\omega} & D_{11,i}^T \\ (C_{1,i} X + D_{12,i} Y_j)^T & D_{11,i} & -\gamma I_{nz} \end{bmatrix} < 0 \\ X > 0 \\ i = 1, \dots, l \\ j = 1, \dots, l \end{cases} \quad (20)$$

Where:

- $X = X^T \in R^{(n+p) \times (n+p)}$.
- $Y_j \in R^{m \times (n+p)}$.
- n is the number of states.
- m is the number of inputs.
- p is the number of output.
- $n\omega$ is the number of state variables that we want to optimize.
- $n\omega$ is the number of disturbances acting on the system.
- l is the number of working points.

if a solution exists then:

1. $(A(p) + B_1(p))K$ is robustly quadratically stable.
2. $\|T_\infty\|_{H_\infty}^{WC} < \gamma$.

3. $\gamma \in R$ is the minimum upper bound for $\|T_\infty\|_{H_\infty}^{WC}$.

The optimal controller is given by:

$$K_{\infty.LPV}(p)^* = \sum_j^l p_j(t) Y_j^* (X^*)^{-1} \quad (21)$$

As introduced, the longitudinal velocity is the parameter on which one intends to perform gain scheduling. Its range of uncertainty is:

$$10 \left[\frac{m}{s} \right] \leq V_x \leq 25 \left[\frac{m}{s} \right] \quad (22)$$

More in details, the following operating ranges are defined:

$$V_x^{(j)} \in \begin{cases} [10 \ 15], & j = 1 \\ (15 \ 25], & j = 2 \end{cases} \quad (23)$$

Thus, three controllers are interpolated to obtain the final gain. Solving the optimization problem with the following performance output:

$$z_\infty = \begin{bmatrix} 7 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_e \\ \beta \\ \dot{\psi} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 6377.4 \end{bmatrix} \delta + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \omega \\ \dot{\psi}_{des} \end{bmatrix} \quad (24)$$

The optimal gains are:

$$\begin{aligned} K_{1,i} &= [169.8508 \quad -15.2673 \quad -12.7668] \\ K_{2,i} &= [166.5915 \quad -14.9660 \quad -12.5169] \\ K_{3,i} &= [162.0954 \quad -14.5499 \quad -12.1721] \end{aligned} \quad (25)$$

Making use of the Automated Driving Toolbox in Simulink, Figure 5 shows the implementation of the closed loop system:

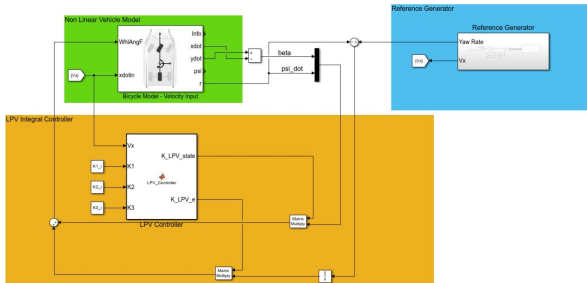


Figure 5: *Simulink Implementation*

Three main systems can be identified:

1. **Non Linear Vehicle Model:** it implements the dynamic of the bicycle model taking into account also non linear factors regarding the interaction tire-road.
2. **Reference Generator:**

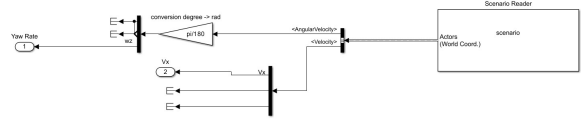


Figure 6: *Reference Generator*

It provides the reference signals in terms of longitudinal velocity V_x and yaw rate $\dot{\psi}_{des}$. These information are captured from the evolution of the following *Driving Scenario Designer*:

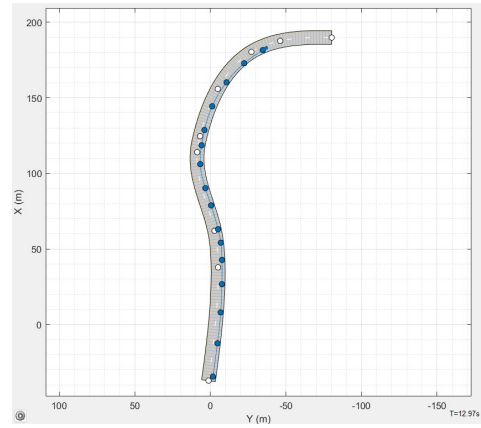


Figure 7: *Designed Trajectory*

Which presumes that the vehicle must accomplish several cornering manoeuvres at different speeds.

3. **LPV Integral Controller:** it is responsible for implementing the interpolation of the *local* controllers and for providing K_{LPV_state} and K_{LPV_e} which respectively are the gains for the state $[\beta, \psi]^T$ and the integrator state $[x_e]$. Then, the input is computed by the following state feedback:

$$u = \delta = K_{LPV_e} * [x_e] + K_{LPV_state} \begin{bmatrix} \beta \\ \psi \end{bmatrix} \quad (26)$$

Since the bicycle model must be fed with the longitudinal velocity in order to simulate its dynamic, the vehicle speed $\dot{x}$ and the reference velocity V_x are equal. Thus, the latter is also received by the function to perform gain scheduling in three steps:

1. First, the current value of the speed is used to choose the correct operating range according to 23. Then, the normalized X variable in $[0, 1]$ is created through the following formula:

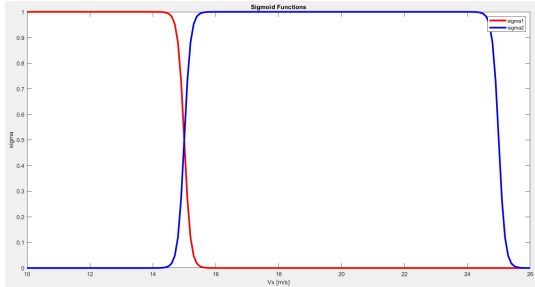
$$X = \frac{V_x(t) - \underline{V}_x}{\overline{V}_x - \underline{V}_x} \quad (27)$$

With $\underline{V}_x, \overline{V}_x$ denoting respectively the minimum and maximum value of the speed in the operating range.

2. Sigmoid functions are created to allow smooth transitions between operating ranges. In particular they are:

$$\begin{aligned} \sigma_1(V_x) &= 1 - \frac{1}{1 + e^{-b(V_x - 15)}} \\ \sigma_2(V_x) &= \frac{1}{1 + e^{-b(V_x - 15)}} - \frac{1}{1 + e^{-b(V_x - 25)}} \end{aligned} \quad (28)$$

With $b = 10$ their evolution is:



3. The controller K_{LPV} is computed according to:

$$K_{LPV} = \sum_{j=1}^l \alpha_j(X) K_j \quad (29)$$

Where:

$$\alpha_j(X) = \sigma_{j-1}X + \sigma_j(1 - X) \quad (30)$$

with: $\sigma_0 = 0, \sigma_l = 0$. Since the LPV controller must be a convex combination of the *local* controllers let's check some properties:

- $\alpha_j(X) \in [0, 1]$. True since $X \in [0, 1]$ by definition and σ_j are functions limited in $[0, 1]$ as seen in 2.
- $\sum_{j=1}^l \alpha_j(X) = 1$. True since:

$$\begin{aligned} \sum_{j=1}^l \alpha_j(X) &= \sum_{j=1}^l \sigma_{j-1}X + \sigma_j(1 - X) = \\ &= X \sum_{j=1}^l \sigma_{j-1} + (1 - X) \sum_{j=1}^l \sigma_j \stackrel{\sigma_0=0}{=} \\ &\stackrel{\sigma_l=0}{=} (X + 1 - X) \sum_{j=1}^l \sigma_j = \sum_{j=1}^l \sigma_j \end{aligned} \quad (31)$$

Thus, we have obtained:

$$\sum_{j=1}^l \alpha_j(X) = \sum_{j=1}^l \sigma_j(X) \quad (32)$$

And we know that is equal to 1 from the structure of the sigmoid functions.

4 Simulation

The simulation accounts the trajectory 7. From it, the following velocity profile is determined:

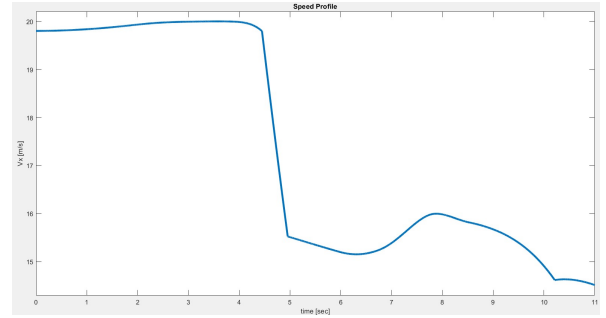


Figure 8: *Speed Profile*

The state evolution of the closed-loop augmented system is here reported:

β Evolution

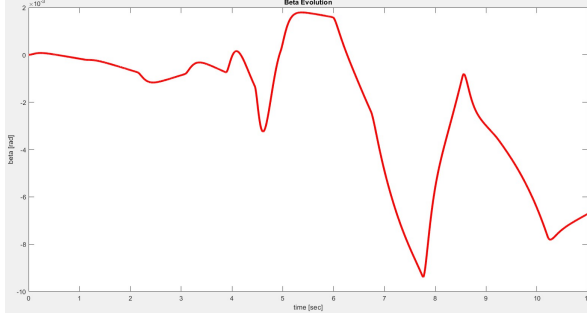


Figure 9: $x_2 = \beta$

Yaw Rate Tracking

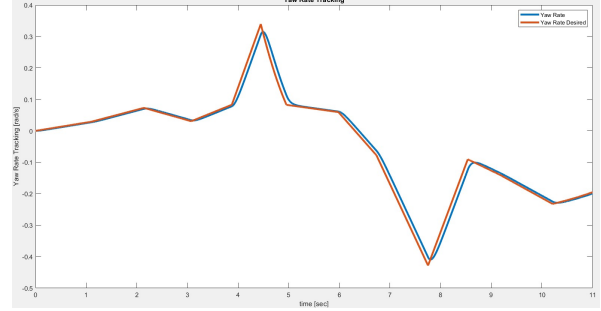


Figure 12: Tracking of $\dot{\psi}_{des}$

Yaw Rate Tracking Error

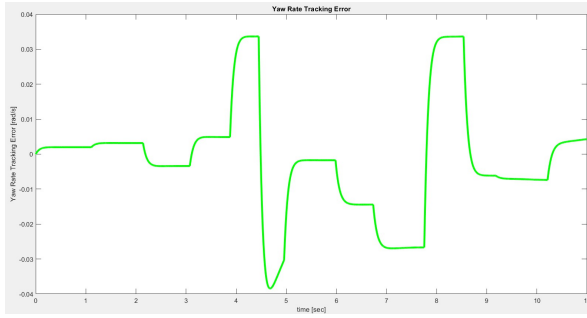


Figure 10: $x_1 = \dot{\psi}_{des} - \dot{\psi}$

The following table summarizes the main results of the simulation:

Variable	Value	Unit
β	$ \beta(t) \leq 0.00938$	rad
$\dot{\psi}_{des} - \dot{\psi}$	$ x_e(t) \leq 0.038$	$\frac{rad}{s}$
δ	$ \delta(t) \leq 0.077$	rad

It is reminded that the control objectives were:

- regulation of the sideslip angle β .
- tracking of the reference $\dot{\psi}_{des}$.
- limited usage of the steering wheel angle δ to avoid sudden variations.

Delta Evolution

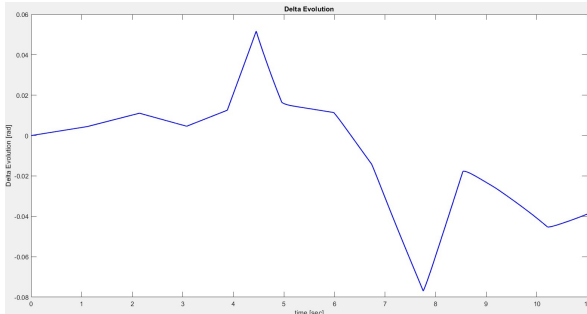


Figure 11: δ

It is possible to make a comparison between the desired yaw rate trajectory and the simulated one:

Looking at the above table is possible to state that the obtained results are appreciable.

Final simulations regards the introduction of the disturbance effect. In the optimization problem formulated only a disturbance acting on the sideslip angle was considered. This effect may be caused by sudden gusts of wind or bank angles of the road, as described in the model 7. Since the inner equations of the non linear Simulink bicycle model are not accessible, to simulate a variation of the road profile, a step signal of 10° is added to the steering wheel starting from 8 [sec]. This scenario aims to analyse the behaviour of the vehicle on a road which is not flat. As a consequence, changes of the bank angle determine changes of the input (thus producing effects on the sideslip angle). The following figures report the results obtained:

β Evolution

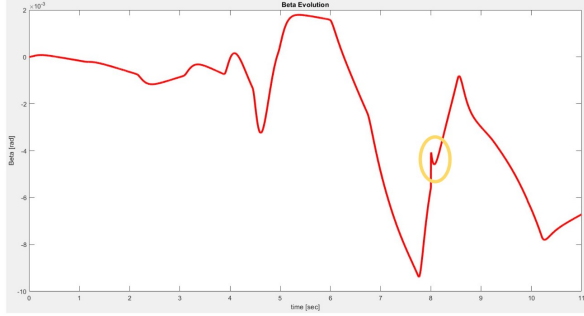


Figure 13: Step Disturbance effect on $x_2 = \beta$

Yaw Rate Tracking Error

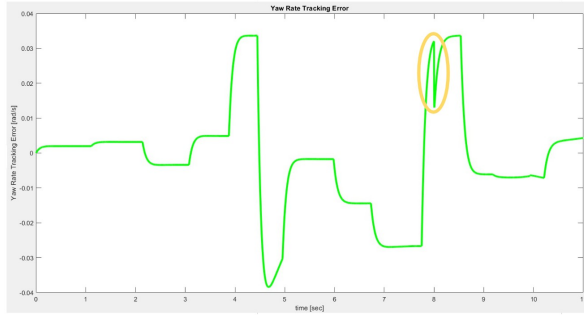


Figure 14: Step Disturbance effect on $x_1 = \dot{\psi}_{des} - \dot{\psi}$

Delta Evolution

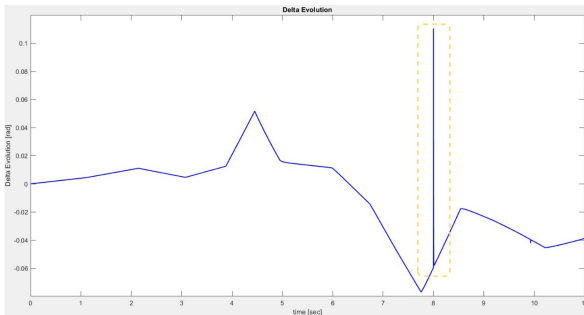


Figure 15: Step Disturbance effect on δ

Yaw Rate Tracking

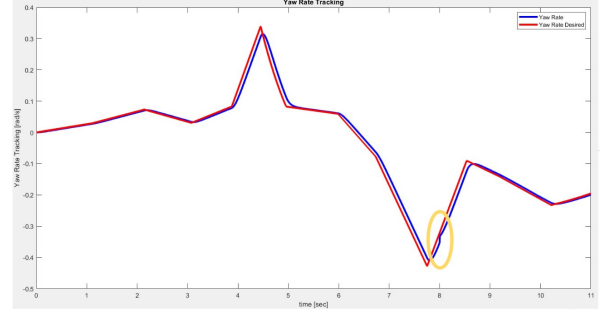


Figure 16: Step Disturbance effect on Tracking of $\dot{\psi}_{des}$

In the above figures the moment when the disturbance arises is highlighted. It is possible to appreciate that, despite an expected initial peak, in presence of a constant step disturbance, the system correctly rejected the disturbance.

5 Conclusions

Summing up, the work aimed to simulate the lateral dynamic of a car having as objectives the stability of the vehicle and the tracking of the yaw rate along a designed trajectory. The simulations involved the Automated Driving Toolbox in which an H_∞ LPV controller was implemented. This strategy has successfully sorted off the problem also in presence of disturbance guaranteeing: perfect rejection, regulation, tracking and limited usage of the steering wheel input.

6 References

- [1] A.Casavola, Notes for Vehicle Control Course.
- [2] G.Gagliardi, Exercices for Vehicle Control Course.
- [3] Automated Driving Toolbox Documentation, Mathworks